

Executive Summary: Telco Customer Churn Analysis

This analysis provides a comprehensive overview of churn behavior among customers of a telecom company. Using effective data visualization and exploration, the key churn drivers were identified and quantified. The insights below are derived from demographic, contractual, and service-based attributes in the dataset.

Key Insights:

1. Overall Churn Rate

- From the **pie chart**, it was observed that **26.54%** of customers have **churned**, while **73.46%** have **remained** with the company.
 - This highlights that **over a quarter** of the customer base is being lost, indicating a need for urgent retention strategies.
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2. Churn by Gender

- Distribution of customers by gender:
 - **Male:** ~50.5%
 - **Female:** ~49.5%
 - Churn patterns are **nearly identical across both genders**, showing **no significant correlation** between gender and churn behavior.
 - **Implication:** Gender-based targeting is unlikely to yield high impact in churn reduction.
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3. Churn Among Senior Citizens

- Senior citizens make up **16.2%** of the total customer base.
- **Churn Rate by Group:**
 - **Senior Citizens:** **41.7%** churned
 - **Non-Senior Citizens:** **24.0%** churned
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- **Insight:** Senior citizens are **73.7% more likely** to churn than non-seniors.
 - **Recommendation:** Consider specialized support or plans targeting senior citizens to reduce this high churn rate.
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4. Tenure vs Churn

- A histogram revealed that churn is **heavily concentrated among customers with low tenure**:
 - Customers with **tenure < 6 months** show a churn rate of **nearly 60%**.
 - Conversely, customers with **tenure > 30 months** show churn rates below **10%**.
 - **Insight:** Customers tend to leave early if they are not satisfied.
 - **Recommendation:** Implement a strong **onboarding program** and **early engagement strategy** for new users.
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5. Contract Type and Churn

- **Churn by Contract Type**:
 - **Month-to-month: 42.0%** churned
 - **One-year: 11.5%** churned
 - **Two-year: 2.8%** churned
 - **Insight:** Longer contracts lead to significantly lower churn.
 - **Recommendation:** Provide **discounts or value-added incentives** to encourage customers to switch from month-to-month to longer contracts.
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6. Churn Across Services

From subplot charts across services like internet, tech support, and streaming, the following patterns were observed:

Service Feature	Churn Rate with Service	Churn Rate without Service
Online Security	15%	45%
Tech Support	17%	40%
Online Backup	20%	38%
Streaming TV	28%	35%

- **Insight:** Customers who **use more services** (especially OnlineSecurity, TechSupport) are **less likely to churn**.
- **Recommendation:** Promote **bundled services** and educate customers about available service features.

Strategic Recommendations

1. **Onboarding & Retention:** Target new users (0–6 months) with onboarding programs, discounts, and customer success outreach.
2. **Contract Optimization:** Encourage upgrades to 1- or 2-year contracts with loyalty benefits.
3. **Senior Citizen Plans:** Design simplified and affordable plans tailored for senior customers.
4. **Service Adoption Campaign:** Promote awareness and usage of features like Online Security and Tech Support.